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People on the move: exploring the functional roles of deprived neighbourhoods

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Abstract. Given the neighbourhood focus of much regeneration policy, we need to know more about the functional roles that neighbourhoods play in the way that households move within the housing market and hence about the different functional types of neighbourhood amongst deprived areas. Such knowledge would help both to guide the priorities of policy and to interpret the probability of policy interventions being successful. This exploratory study draws on an evaluation of the British government's National Strategy for Neighbourhood Renewal, part of which entails an interpretation of household mobility data from the 2001 Census. It suggests four categories of neighbourhood—transit, escalator, isolate, and improver areas—based on the relationship between where households move to and move from, focused on the 20% most deprived lower super output areas in England. Evidence on the ground suggests the plausibility of the different functional roles played by the four neighbourhood types. Some continuing conundrums—the robustness of the categorisation, the need to take account of the spatial context of deprived areas, and the difference between movers and stayers—are discussed as a prelude to further continuing research.

In developing its regeneration policy the British government has put great weight on the role of the neighbourhood. Yet, despite the policy focus, and the huge literature, on neighbourhoods, in some important respects we know rather little about the functional roles that neighbourhoods play; nor do we have very robust typologies of different functional types of deprived neighbourhood. If policy is to be targeted more effectively at neighbourhoods, it seems important that we should understand both the roles that neighbourhoods play and the range of neighbourhood types at which policy is addressed.

The huge literature on neighbourhoods is, of course, an international one that transcends disciplinary boundaries and methodological approaches. It has largely focused on three issues: neighbourhood composition, area effects, and the concept of the mixed neighbourhood. One strand of work—on neighbourhood composition—has generally adopted traditional approaches which seek to categorise areas on the basis of internal composition. In the UK such methods generally stem from the socioeconomic classification methods adopted by Webber and Craig (1978) and from subsequent commercially developed geodemographic systems (see Sleight, 2004). There has recently been a move to diversify such population segmentation techniques (Vickers and Rees, 2007), but it remains the case that market-driven classifications currently dominate the field, at least in terms of utilisation and application, despite wider European and American contributions to the literature on neighbourhoods and social composition (eg Friedrichs et al, 2003). Unsurprisingly, given the commercial and essentially descriptive nature of neighbourhood composition approaches, it remains an area of debate in scholarly circles (eg Voas and Williamson, 2001).

A clear picture of the technical and conceptual breadth of work in the second field—area or neighbourhood effects—is evident from Galster's (1986; 2001) seminal work on the nature of neighbourhood and its effect on different outcomes, Sampson's (2003) research on the neighbourhood context, and Morenoff's (2003) in-depth analysis of the neighbourhood effects paradigm on birth weight. As Lupton (2003a) notes, however, the tendency in the literature has been towards traditional 'community study' approaches rather than more quantitative research which has to some extent been constrained by lack of available data. Dietz (2002) also comments on this weighting towards qualitative research on area effects. The interest in area effects has been sustained over a number of years and it is only recently that technically sophisticated approaches have been able to offer more penetrating insights of the kind illustrated in a recent special issue of *Housing Studies*, which considers the issue of quantifying neighbourhood effects from a variety of perspectives. The range of international authors (eg Andersson et al, 2007; Blasius et al, 2007; Gijsberts and Dagevos, 2007; Oberwittler, 2007) and the variety of topics (such as household income, housing tenure, child poverty, perception of deviance, integration of ethnic minorities) are indicative of a field that is both fertile and vibrant.

Such work on neighbourhood composition has in part fed into the policy interest in mixed neighbourhoods since there have been claims that evidence on area effects supports the development of mixed communities as a means of tackling area deprivation and improving the life chances of deprived households (Fitzpatrick, 2004). In the US, for example, housing programmes offer incentives to encourage better-off households to move into deprived areas (for example, the Nehemiah Housing Opportunity Grants Program) or to move households from deprived neighbourhoods to better neighbourhoods [for example the Moving to Opportunity (MTO) programme]. In Europe, better-off households are encouraged to move to deprived areas through revitalisation programmes (Meen et al, 2005). For example, in the UK, government policy seeks to create mixed communities through mixed housing and tenures that aim to accommodate the needs of different household types and income groups (ODPM, 2003). Such policies, however, are not based on any evidence demonstrating that deprived household's life chances are better in socially mixed communities. In the US, for example, recent findings from studies on the MTO programme found no evidence of positive changes in the economic situation of adults (Cheshire, 2007) while, in the UK, studies have suggested that mixed communities "may produce conflict rather than a sense of community, either because they are enforced or because the British are now much more familiar with homogenous communities" (Kearns and Parkes, 2003, page 847).

The principal gap in this literature is the functional role that neighbourhoods play in the sorting mechanism of different households. Ironically, this concern about the dynamics of neighbourhoods has a long lineage. It was addressed in some of the work of the 1920s' Chicago School of urban sociologists through the work of Park and Burgess (1925) in their identification of 'zones of transition' and in Hoyt's (1939) work on the spatial dynamics of high-cost housing. The contribution of what follows in this paper, therefore, is not so much as a new piece of evidence for the area effects literature but, rather, an attempt to fill a gap which currently exists in relation to the different roles played by neighbourhoods in a spatially dynamic context. In this way, we aim to add to the evidence base on the importance of the neighbourhood as the context for targeted policy. By taking an approach to understanding neighbourhoods that is based on the use of data on household mobility rather than on the more traditional cross-sectional 'ecological' data, this paper seeks to offer something relevant beyond the English context within which it has been developed.

The policy context

The centrality of the neighbourhood within British government policy is in part manifested through the growth of area-based initiatives such as the New Deal for Communities (NDC) programme, Neighbourhood Wardens, and Neighbourhood Management Pathfinders. The current National Strategy for Neighbourhood Renewal set out the government's two long-term goals: to improve conditions in all the poorest neighbourhoods in terms of worklessness, housing, crime, health, and skills; and to narrow the gap between the most deprived neighbourhoods and the rest (SEU, 2001). Such programmes aim to tackle inequalities at an area level in the belief that some of these inequalities may be self-reinforcing (McCulloch, 2001) and that people's circumstances may be adversely affected by the nature of the area in which they live—the 'neighbourhood effect' thesis (eg Lupton, 2003a; 2003b). Almost invariably, the selection of small areas targeted for such interventions has been guided by measures of deprivation such as the Index of Multiple Deprivation (IMD). However, neighbourhood effects can exist at different levels and it is the interaction between these that shape them. It is therefore important to know something about the wider context within which a neighbourhood sits as well as the composition of the neighbourhood's own households. But it is equally important to understand something about the role that a neighbourhood plays in the intersection of the labour and housing markets. Each of these three aspects can have important implications both for the types of policy intervention that may be most relevant and for interpreting the probability that interventions will be successful in their aim within specific types of area.

Context is significant because it matters whether a deprived neighbourhood is or is not surrounded by or close to more prosperous areas. For example, where a deprived area sits in a broader area that is economically buoyant, the challenge for regeneration is usually to achieve a better linkage between the deprived neighbourhood and its wider labour market. This may be a question of improving physical links, or of improving the skills base of the local population, or of tackling perceptions both of the local population and of potential employers. On the other hand, where deprived neighbourhoods sit within wider areas of depressed economies, the challenge is more likely to be one of boosting demand across the broader area. In terms of the likelihood of success, the former would appear the easier of such challenges.

The importance of neighbourhood *composition* lies in helping to guide the priorities that policy might best address. Deprived areas with a predominantly elderly population will have very different priorities from an area that is largely composed of families with young children. And, even though many of the indicators of deprivation are highly correlated, neighbourhoods with exceptionally high crime rates, for example, would be likely to have different priorities from those with poor health; those with a uniform housing tenure would differ from those with more mixed housing composition.

Such differences are now reasonably well understood. The great bulk of literature on the neighbourhood has focused on such issues, concentrating on aspects of neighbourhood effects, the socioeconomic composition of neighbourhoods, and the implications and feasibility of developing mixed communities. However, the *roles* that neighbourhoods play in the flux of household flows within the housing market are rather less well explored. This is the focus of what follows. Can we, by looking at the residential churn and the characteristics of households moving into and out of a neighbourhood, begin to develop a better understanding of the roles that different neighbourhoods play? Can we outline a typology of neighbourhoods based on such roles? Answering such questions would both help to determine the best focus of policy interventions and affect our interpretation of how successful those interventions have been.

Exploring residential mobility

Residential churn can clearly impact upon deprived neighbourhoods in different ways. Incoming disadvantaged migrants can destabilise existing communities by threatening social networks, creating problems of crime and vandalism, and putting pressure on local public services. Conversely, if in-migrants are less deprived or less fatalistic or less socially isolated than existing residents, residential churn could change the existing social mix and improve the area as a whole. This can constitute one form of gentrification. The in-migration of wealthier individuals into deprived areas may be due to housing-market pressures—as in inner parts of cities such as Shoreditch in London, which has attracted professionals wishing to live close to the City (SEU, 2004). Often accompanied by investments in the physical environment, the influx of better-off home-buyers and renters eventually culminates in neighbourhood 'refurbishment' or renewal (Smith, 1996).

The flow of households into and out of areas also can play an important role in affecting the nature of social networks which can pass on information about job opportunities (SEU, 2004). The assumption here is that individuals living in well-connected deprived areas would have better access to external opportunities than those living in isolated deprived areas. The extent to which deprived areas have migration interactions with other deprived areas is an important topic, since any change in the mix of population will result in changes in the profiles of areas over time (O'Reilly and Stevenson, 2003).

In looking at such issues, the 2001 Census is an invaluable source for exploring the inflows and outflows of households.(1) It provided the base for Bailey and Livingstone's (2007) authoritative and exhaustive recent study of migration. In it they distinguish three aspects of migration as it impinges on neighbourhoods: area stability, area connectivity, and area change. The most intriguing aspect of migration is what Bailey and Livingstone call 'connectivity'—the types of area from which and to which households move. For regeneration policy this has two significant implications. First is the impact of any asymmetry between the people moving into and those moving out of a deprived neighbourhood. If 'those who can, move out', any improvements to the circumstances of these individuals will not be reflected in improvements to the deprived areas from which they come. For example, if unemployed people move out of a deprived area once they get a job, and are replaced by in-movers who are worse-off (in this case, not employed), this will mean that the benefits to individuals will not be matched by benefits to areas. The evaluation of the NDC programme produced evidence of this tendency (CRESR, 2007) on the basis of successive household surveys taken at two-year intervals. They found that those who moved out of NDC areas were more likely to be employed and have higher educational attainment than those who moved in.(2) Where such asymmetry does occur it clearly implies that any formal evaluation of the area-based impacts of policy interventions will understate the benefits that accrue to individuals.

Second is the comparison between the levels of deprivation of the neighbourhoods between which households move. This is the focus of what follows. Exploring these links may begin to throw some light on the different roles that neighbourhoods play in the housing and labour markets. By comparing the levels of deprivation of the neighbourhoods from which and to which households move we can hypothesise a variety of types of links between deprived areas and those with which they are connected by

(1) The analysis that follows would have been more robust were there longitudinal data for England of the kind that exist, for example, in The Netherlands (Musterd et al, 2003).

(2) However, because of the difficulty of following up households, it proved possible to derive data for only 300 households across the 39 NDC areas, hence the data inevitably lack a degree of robustness.

	Less deprived	Same	More deprived
Transit			
In-movers	•		
Out-movers	•		
Escalator			
In-movers		•	•
Out-movers	•		
Isolate			
In-movers		•	•
Out-movers		•	•
Improver			
In-movers	•		
Out-movers		•	•

Figure 1. A typology of deprived areas.

migration flows. These linkages can provide the basis for an exploratory definition of the different roles played by deprived areas. Figure 1 suggests four ideal types of role that deprived neighbourhoods might play based on this relationship between the deprivation scores of the neighbourhood in question and that of the neighbourhoods from which in-movers come and out-movers go. The cells with the largest flows are indicated by a dot. This results in the pattern of flows shown in figure 2.

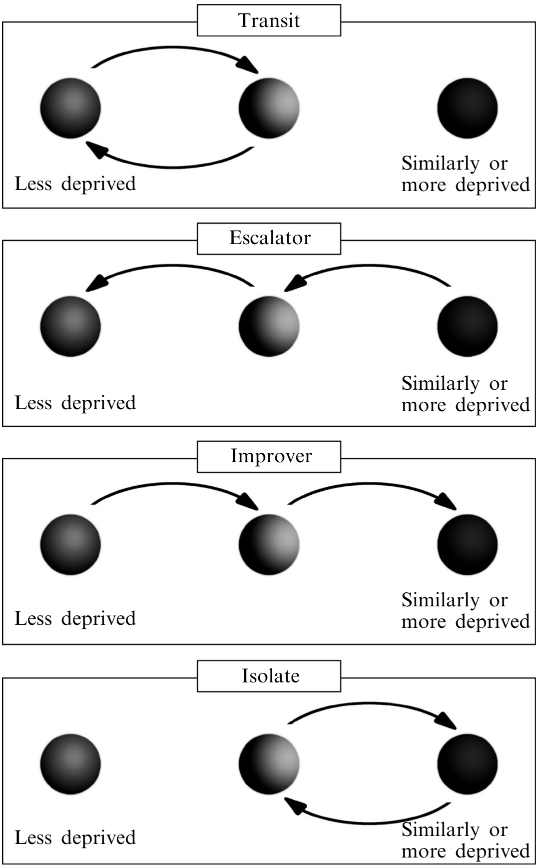


Figure 2. Patterns of flows between neighbourhoods.

Transit areas are deprived neighbourhoods in which most in-movers come from less deprived areas and most out-movers go to less deprived areas. Typically, this implies young or newly established households coming from more 'comfortable' backgrounds and starting out on the housing ladder. Their early choice of housing—and hence location—reflects their initially limited resources. For them, living in a deprived neighbourhood may entail only a short period of residence before they move elsewhere to a 'better' area.
Escalator areas play a not dissimilar role, but in their case, since most of the in-movers come from areas that are equally or more deprived, the neighbourhood becomes part of a continuous onward-and-upward progression through the housing and labour markets. The moving households may be older than those in the transit areas since they would not necessarily be at the start of their housing career.
Isolate areas represent neighbourhoods in which households come from and move to areas that are equally or more deprived. To this degree, they are neighbourhoods that are associated with a degree of entrapment of poor households who are unable to break out of living in deprived areas.
Improver areas are ones in which there is a degree of social improvement since most in-movers come from less deprived areas. This could be seen as a form of gentrification. However, it may or may not entail the kind of conscious process of markedly richer households displacing markedly poorer households envisaged by much of the literature that discusses gentrification (see Lees, 2000). Hence, we prefer to use the term 'improver' rather than 'gentrifier'.

If deprived neighbourhoods play one or other of these four roles there are clearly implications for policy interventions. For example, for those moving through the transit and escalator neighbourhoods, the areas can be argued to play an important functional role within the broader housing market since, despite their level of deprivation, the neighbourhood provides affordable housing for those at a generally early stage of housing progression. Improvements to the stock of housing may in part be expected to come not from the public purse but from private investment. For improver areas, there are policy issues about the need to provide decent housing for those deprived households moving out to other neighbourhoods; hence, one of the questions relevant to regeneration may be the broader housing market outside the immediate targeted deprived neighbourhood itself. For isolate areas, there is perhaps the strongest case for comprehensive policy intervention to improve the prospects of all households. Typically, it is these areas which may be expected to form the heart of areas targeted by comprehensive regeneration programmes such as Housing Market Renewal.

Identifying neighbourhood types

Whether or not such ideal types of neighbourhood exist is, of course, an empirical question on which census data can help to throw some light since the census data record the origin and destination of migrant households during the year 2000/01. To explore this, data on household migration from the census have been used for those lower super output areas (LSOAs)[3] that fall within the worst 20% of the IMD 2004. The focus is restricted to deprived areas since a rather different logic would apply to the roles played by nondeprived neighbourhoods. In total, this entails focusing on 6496 deprived target LSOAs and recording moves to or from any of the 32 482 LSOAs across the country. Households moving into or out of one of the deprived LSOAs can therefore

[3] LSOAs are spatial reporting areas used by the census. They are amalgamations of the most basic census area (output areas) and contain on average some 1513 people.

be attributed the IMD rank of the relevant LSOA from which and to which they moved in order to define each LSOA in terms of the fourfold typology of figure 1.[(4)]

Applying the definitions that are outlined below to the 20% most deprived LSOAs produces 1213 escalator areas, 521 improver areas, 2030 isolate areas, and 2519 transit areas (the remaining 213 deprived LSOAs do not fall unambiguously into any of the four types and were therefore excluded from the analysis). As shown in table 1, there is significant variation in the classification of areas between different types of local authority districts. For example, there is a high percentage of isolate LSOAs in conurbations; high percentages of transit LSOAs in London, large free-standing cities, seaside resorts, and rural areas; higher percentages of improvers in London and in large free-standing cities than in other types of area; and higher percentages of escalators in conurbations, industrial and mining districts, and in inner London. This adds some confidence to the belief that the four neighbourhood roles may be functionally meaningful rather than merely statistical artefacts.

The real test of the meaningfulness of the four categories is to explore their pattern on the ground. Encouragingly, the suggested typology does appear in practice to produce empirical results which make some sense in terms of local knowledge of different areas. This can be illustrated with the patterns produced for three areas: Manchester, Liverpool, and London (figures 3–5).

Table 1. Escalator, improver, isolate, and transit areas by type of district.

Type of local authority	Number of LSOAs[a]				Percentage of LSOAs			
	escalator	improver	isolate	transit	escalator	improver	isolate	transit
Conurbation core	259	111	677	271	19.7	8.4	51.4	20.6
Conurbation industrial	408	150	767	430	23.2	8.5	43.7	24.5
Industrial and mining	126	56	165	226	22.0	9.8	28.8	39.4
Large free-standing city	56	35	63	197	16.0	10.0	17.9	56.1
Large free-standing town	30	17	15	315	8.0	4.5	4.0	83.6
London core	231	107	292	427	21.9	10.1	27.6	40.4
London dormitory	39	11	9	254	12.5	3.5	2.9	81.2
Non-London dormitory	27	15	21	96	17.0	9.4	13.2	60.4
Rural and semirural	23	12	15	184	9.8	5.1	6.4	78.6
Seaside resort	14	7	6	119	9.6	4.8	4.1	81.5
All deprived neighbourhoods	1213	521	2030	2519	19.3	8.3	32.3	40.1

Note: The classification of districts is a modified form of a typology originally used by the Office for National Statistics, which is based on a mixture of the administrative status and the occupational structure of districts.
[a] LSOA—lower super output area.

[(4)] The analysis excludes people moving from or to places outside the UK because they cannot be related to IMD ranks and because international out-movers are not recorded. This is especially unfortunate in the case of London neighbourhoods where international migration has been especially high. The analysis also excludes those with no fixed address.

Manchester and Liverpool suggest interesting similarities and differences (figures 3 and 4). Both have large numbers of isolate areas amongst their deprived LSOAs, but the percentage is much higher in Liverpool, with almost the whole of north Liverpool and significant parts of the south (not least in Speke-Garston) featuring as isolate areas. In contrast, Manchester's isolate areas are far fewer as a percentage of its deprived areas and are found overwhelmingly in the north and east (and in the large local authority estate of Wythenshawe in the south), where indeed many of the regeneration programmes have been concentrated. Both districts show many more transit and escalator neighbourhoods in their southern areas, again conforming to local expectations. Equally, both show improver neighbourhoods around their central areas, where the growth of new and refurbished apartments has reflected the trend to inner-city living. For Manchester, the improver areas also include parts of Hulme (where new-build housing linked to a City Challenge programme has transformed an area that had been dominated by local authority housing), and in parts of Chorlton and Whalley Range in the southwest, where the private housing market has shown rapid growth in prices and popularity.(5)

The London boroughs (figure 5) suggest a very different pattern. Isolate areas are far fewer and are largely restricted to the east end of the city. There is a marked contrast in this respect between some of the eastern boroughs. Newham and Hackney, on one hand, have a relatively high concentration of isolate neighbourhoods, whereas Tower Hamlets has relatively few. In the latter (and in parts of Hackney and Newham that lie close to the city), the pressures of the London housing market (and the influence of gentrification) have helped to transform the nature of many erstwhile unpopular inner-area locations. In contrast to Manchester and Liverpool, far more of the London LSOAs are transit neighbourhoods where young households start out on the housing (and career) ladder, again reflecting the particular pressures of the London housing market. Interestingly—and unsurprisingly—many of the east London neighbourhoods which are not isolates fall into the escalator rather than the transit category, suggesting that these are areas where poorer households are in the process of upgrading their position in the housing market. London also shows a relatively large number of improver areas—some of which are in the classic 'gentrifying' areas like Islington. Such differences clearly reflect the very distinctive nature of London's housing market and its severe problems of a lack of affordable houses.

These patterns appear to make much sense of what is known about the respective roles of different neighbourhoods in each of these districts. Despite the fact that all the target LSOAs are amongst the 20% most deprived in the country, it seems clear that they do play different roles in terms of the churn of the housing market (and, by implication, the labour market). It is significant that there is only a very weak relationship between deprivation scores and each of the four neighbourhood types. As table 2 shows, the distribution of IMD scores across the four types is very similar. What small differences do exist are understandable in terms of the neighbourhood roles. For example—as might be expected—the highest average IMD score is for the isolate areas, followed by the escalators areas; and the lowest is for improver areas. However, the differences are very small. What is striking is that more deprived and less deprived LSOAs are found across all four types of neighbourhood, suggesting that the typology is not simply a reflection of the degree of deprivation amongst the set of neighbourhoods but, rather, an aspect of the different functional roles that deprived neighbourhoods play in the housing market.

(5) The improver area in the extreme south gives a misleading impression since the single large LSOA includes Manchester Airport.

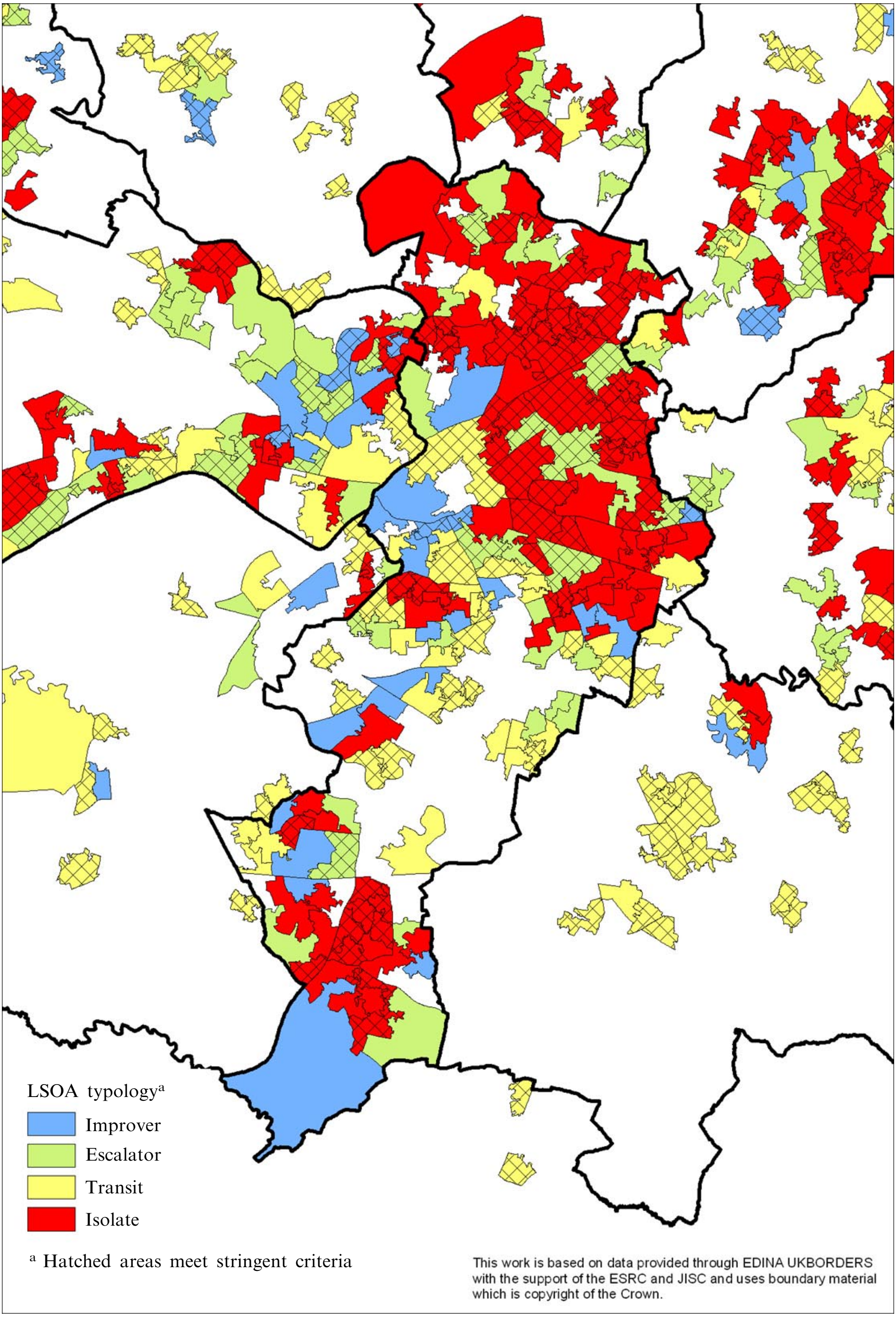


Figure 3. [In colour online, see http://dx.doi.org/10.1068/a40241] Neighbourhood typology: deprived lower super output areas (LSOAs) in Greater Manchester.

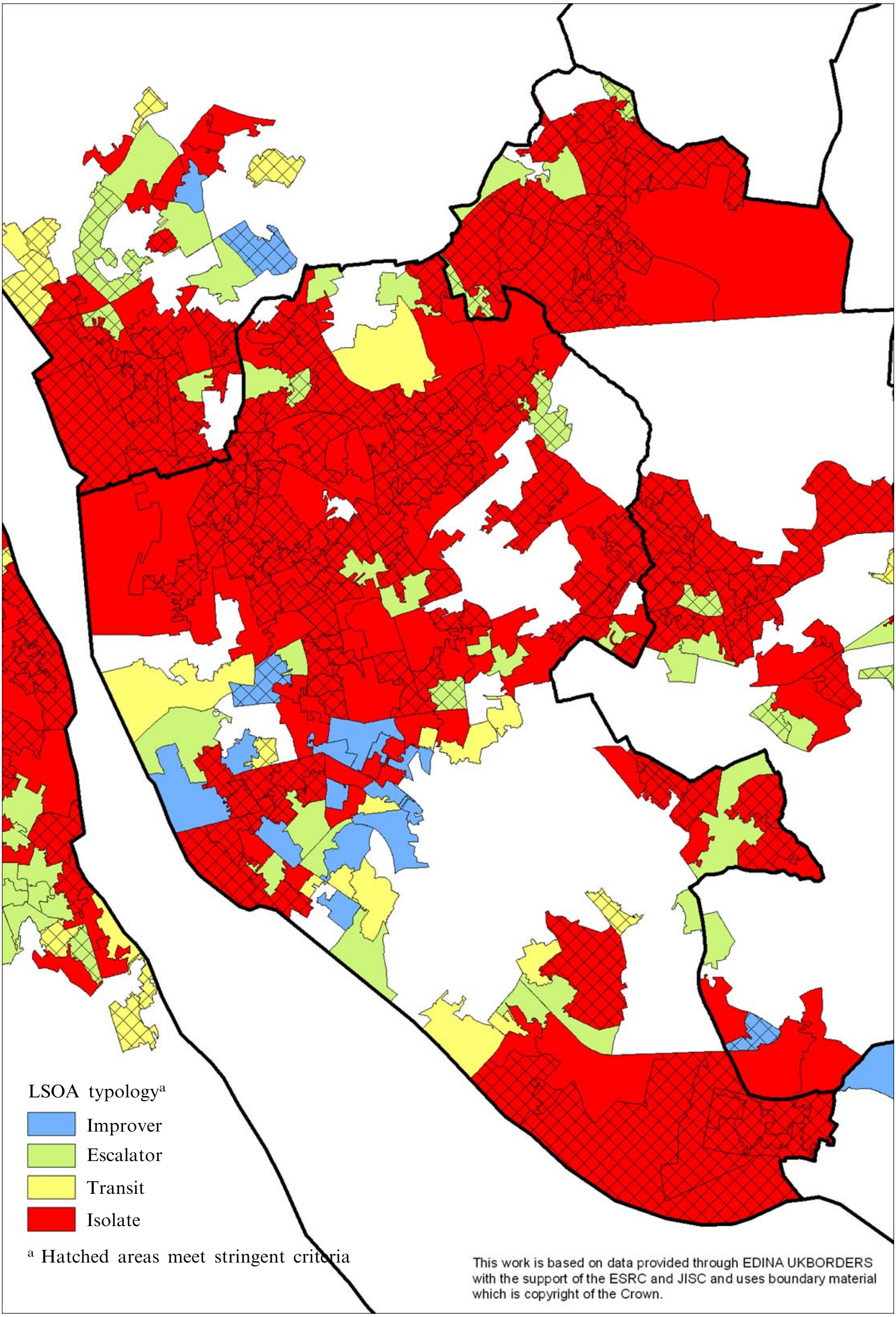


Figure 4. [In colour online.] Neighbourhood typology: deprived lower super output areas (LSOAs) in Merseyside. (Note: the more stringent definition of the four types of area is based on criteria outlined in a later section and summarised in table 4.)

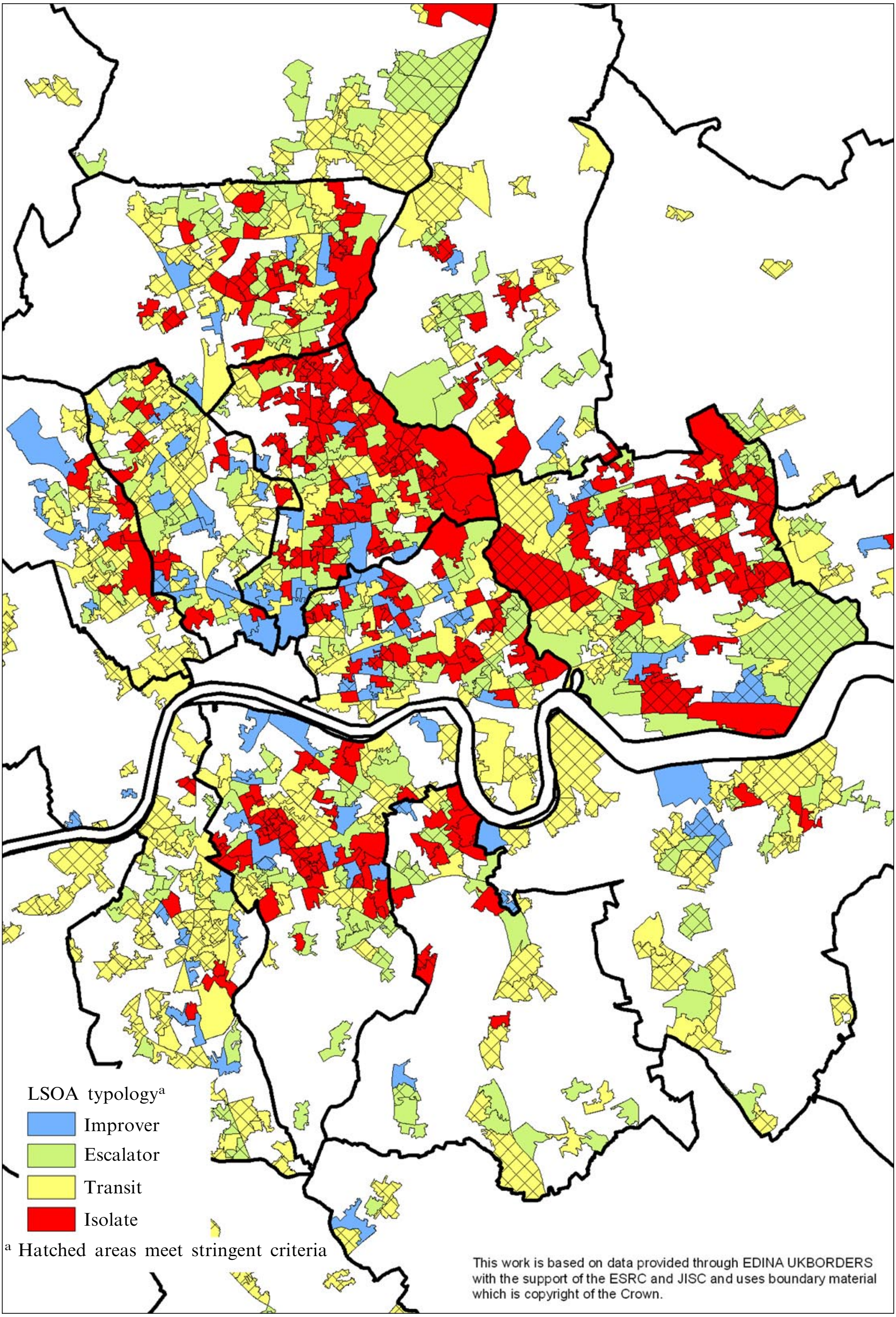


Figure 5. [In colour online.] Neighbourhood typology: deprived lower super output areas (LSOAs) in Greater London (east). (Note: the more stringent definition of the four types of area is based on criteria outlined in a later section and summarised in table 4.)

Table 2. The neighbourhood typology and deprivation.

Area	IMD scores			
	average	standard deviation	maximum	minimum
Transit	45.63	9.22	79.99	34.21
Escalator	46.56	9.87	80.29	34.21
Isolate	51.12	11.88	86.36	34.22
Improver	47.71	10.54	78.88	34.21

Note: The higher the IMD (Index of Multiple Deprivation) score the more deprived is a lower super output area.

Limitations of the methodology

Clearly, the definitions used to produce these results are not without their problems. These include both technical and conceptual difficulties, some of which can satisfactorily be addressed but others present real conundrums:

- To preserve anonymity, the census randomises small numbers for data on LSOAs through a mechanism known as small cell adjustment methodology (Stillwell and Duke-Williams, 2007). Where 3 or fewer people are involved in a move from one LSOA to another, values of 0 to 3 are randomly allocated. There are no simple ways in which to circumvent this limitation of the data.
- Attributing the characteristics of neighbourhoods to individuals runs the risk of committing the ecological fallacy. Ideally, one would need to know the characteristics of the moving households themselves, but this is unrealistic at a national scale since it would require a large national survey of movers. The census represents the only source of detailed national data on the characteristics of moving households. Since LSOAs are very small areas (with an average population size of 1513), their very smallness means that empirically they are generally quite homogeneous areas. Practically, of course, the output area is the smallest area for which census origin–destination data are available, but the very small number of movers at output area level makes this too small a scale at which to derive robust data.
- The definition of 'more', 'less', and 'similarly' deprived is clearly not unproblematic. The IMD scores of the 'deprived' LSOAs are the tail end of a normal distribution so, whether one uses differences in values or differences in rank, the interpretation of the significance of a difference will depend in part on where an LSOA falls within the distribution. For LSOAs with high deprivation scores, there will potentially be a higher probability of moving to a less deprived area simply as a result of the skewed distribution. At the extreme, households in the most deprived LSOA can only move to somewhere which is more or similarly deprived.
- Further, the width of the band that is defined as being the 'same' should ideally take account of the number of deprived LSOAs in the proximate areas. This would entail using different band widths for different contexts. However, this would produce too variable an approach. Here, after experimenting with a range of alternatives, we have opted to use a standard value of 10% either side of an LSOA's IMD rank as the definition of 'same'.[6] This appears to produce the most realistic and consistent outcome across the country.

[6] So, for an LSOA ranked 5000 (out of the total of 32 482 LSOAs in England), LSOAs with the same level of deprivation are taken as places between ranks 1752 and 8248.

- There is a related problem in defining into which category the majority of moves fall since the numbers of moving households are sometimes quite small and there are many LSOAs where there are only marginal differences between the number of households moving to or from better, same, or worse areas.
- The context of neighbourhoods can be an important determinant of the probability of households moving to areas that are more or less deprived. If, as is the case, most moves are short distance then whether or not a deprived neighbourhood is close to other similarly (or more) deprived areas would influence the likelihood of moves to similar or more deprived areas. Again, the use of LSOAs is likely to minimise this problem since their small size means that in most local authority districts or wards there is a relatively wide spread of IMD ranks amongst their LSOAs.
- Characterising neighbourhoods by focusing only on migrant households ignores the (invariably) larger proportion of 'stayers' whose characteristics may be very different. It could well be, for example, that transit areas may represent a staging post on the housing ladder for a subset of nondeprived households in a neighbourhood, but that many other households are trapped in the area (as in an isolate area) and have deprivation-related needs that justify policy intervention. The policy implication would, of course, be that in such areas there would be particular need to ensure that policy was targeted on genuinely deprived households.

The final three of these issues—the significance of numerical differences, the neighbourhood context, and 'stayers'—warrant further discussion and exploration.

Robustness of the definitions

First is the question of the robustness of the typologies, given the often quite small number of movers at an LSOA scale. To explore this, two categories can be defined; a more stringent and a less stringent definition. Table 3 specifies the criteria on which these have been based. For example, in the case of transit areas (where the criteria are that most in-movers come from less deprived areas and most out-movers go to less deprived areas), the less stringent definition simply entails the number of in-movers from better areas (IB) being greater than the number of in-movers from similar (IS) or worse (IW) areas; and the number of out-movers from better areas (OB) being greater than the number of those from similar (OS) or worse (OW) areas. On the other hand, the stringent definition ensures that there are significant differences between these numbers. Hence, the in-movers from better areas and out-movers to better areas have to be more than 50% of all movers; and the differences between movers to and from better areas and those from similar or worse areas have to be greater than 10% of all movers. The mapping of LSOA types (as in figures 3 to 5) shows the differentiation into these two categories.

Spatial contexts

Second is the problem of spatial context. For example, if a highly deprived area is surrounded by other similarly deprived areas, the potential for area improvement through migration is more limited, since most moves are of short distance (and particularly so for more deprived households). On the other hand, if a highly deprived area is relatively isolated from other deprived areas then the potential for improvement through migration is in theory much greater. This issue therefore has an impact on the derivation of the typology, particularly in areas with large clusters of highly deprived LSOAs or for more isolated deprived LSOAs.

This broad issue has been widely discussed in recent years, principally in relation to debates on 'area effects' (see Dorling, 2001; O'Reilly and Stevenson, 2003; Smith et al, 2001). In other words, are some areas less likely to have positive outcomes *because* of

Table 3. Definitions of the four typologies (a) notations used; (b) definitions.

	Better (ie less deprived (LSOA)[a]	Same (ie equally deprived LSOA)	Worse (ie more deprived LSOA)
(a)			
Total in-movers	IB	IS	IW
Total out-movers	OB	OS	OW

Area	Stringent definition	Relaxed definition
(b)		
Transit	IB > 50% of all in-movers and OB > 50% of all out-movers and	(IB > IS and IB > IW) and (OB > OS and OB > OW)
	(IB – IS > 10% of all in-movers) and (IB – IW > 10% of all out-movers) and	
	(OB – OS > 10% of all out-movers) and (OB-OW > 10% of all out-movers)	
Escalator	(IW + IS > 60% of all in-movers) and (OB > 50% of all out-movers) and	(IS > IB and IS > IW) and (OB > OW and OB > OS) or
	(IW – IB > 10% of all in-movers) and	
	(OB – OS > 10% of all out-movers) and (OB – OW > 10% of all out-movers) or	(IW > IB and IW > IS) and (OB > OW and OB > OS)
	(IS – IB > 10% of all in-movers) and (OB – OS > 10% of all out-movers) and (OB – OW > 10% of all out-movers)	
Isolate	(IS + IW > 60% of all in-movers) and (OS + OW > 60% of all out-movers) and	(IS > IB and IS > IW) and (OS > OB and OS > OW) or
	(IS – IB > 10% of all in-movers) and (OS – OB > 10% of all out-movers) or	(IS > IB and IS > IW) and (OW > OB and OW > OS) or
	(IS – IB > 10% of all in-movers) and (OW – OB > 10% of all out-movers) or	(IW > IB and IW > IS) and (OS > OB and OS > OW) or
	(IW – IB > 10% of all in-movers) and (OS – OB > 10% of all out-movers) or (IW – IB > 10% of all in-movers) and (OW – OB > 10% of all out-movers)	(IW > IB and IW > IS) and (OW > OB and OW > OS)
Improver	IB > 50% of all in-movers) and (OS + OW > 60% of all out-movers) and	(IB > IS and IB > IW) and (OS > OB and OS > OW) or
	(IB – IS > 10% of all in-movers) and (IB – IW > 10 of all in-movers) and (OS – OB > 10% of all out-movers) or	(IB > IS and IB > IW) and (OW > OB and OW > OS)
	(IB-IS > 10% of all in-movers) and (IB – IW > 10% of all in-movers) and (OW – OB > 10% of all out-movers)	

[a] LSOA—lower super output area.

their location next to other deprived areas? The evidence base is still rather incomplete, despite several attempts to answer this question (eg Atkinson and Kintrea, 2001; Burrows and Bradshaw, 2001; McCulloch, 2001). However, there is an emerging consensus that "the context in which the neighbourhood sits is also a very important influence on neighbourhood outcomes" (Atkinson and Kintrea, 2001, page 2295).

Hence, there is a need to take account of the issue of spatial context and the potential impact it has on the typology explored here.

In order to assess more formally the spatial context of the individual LSOAs used to derive the typology, some further spatial statistical analysis was undertaken. This was done by deriving 'nearest-neighbour' deprivation scores for each LSOA in England, based on the concept of first-order contiguity (see figure 6), whereby the score for each LSOA is derived by averaging the scores for all immediately contiguous LSOAs. This calculation was performed with GeoDa software, using standard geographical information system procedures (Anselin, 2003), and provides further spatial context information that individual LSOA scores alone cannot reveal.

The implication of taking a spatial perspective which acknowledges the conditions in surrounding areas is more than purely conceptual; it can alter our view of spatial structures and shed further light on the potential for positive spatial interaction. For example, the IMD assigns an individual deprivation score to each LSOA in England. This is typically reported in isolation, with no reference to the spatial context of surrounding LSOAs. If we are to acknowledge the possibility of area effects then it makes more sense to consider surrounding areas as a potentially important local influence, particularly in relation to migration. The difference between these two perspectives is demonstrated in figure 7(a) and 7(b), where deprivation scores for north-west England have been mapped using the IMD scores for individual LSOAs [figure 7(a)] compared with the average IMD scores of each LSOA's nearest-neighbour (first-order contiguity) [figure 7(b)].

The IMD scores for individual LSOAs show a more dispersed pattern which highlights concentrations of deprivation in the urban areas but also in some outlying small towns and 'rural' areas. In the map displaying nearest-neighbour scores, the contrast

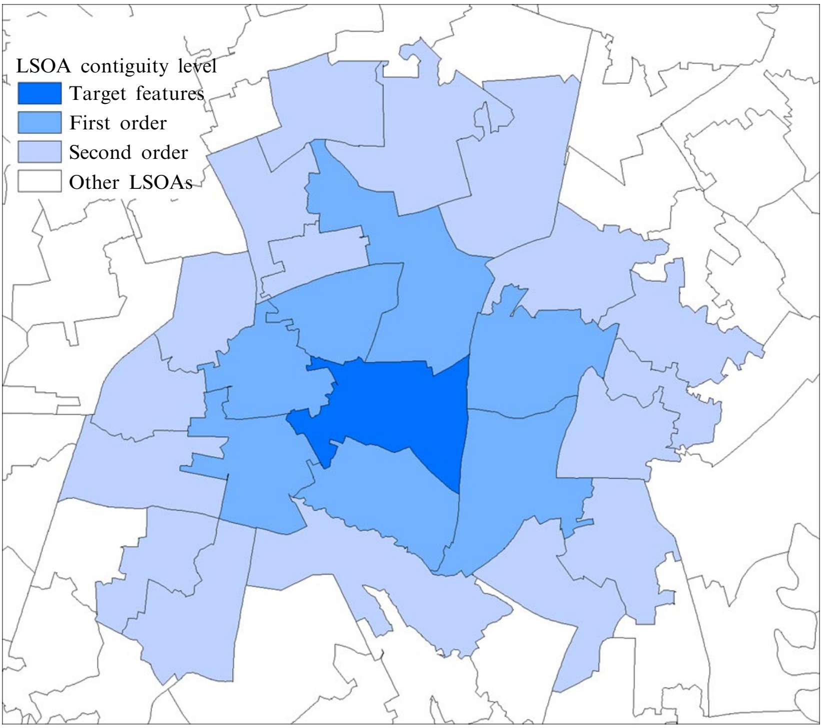
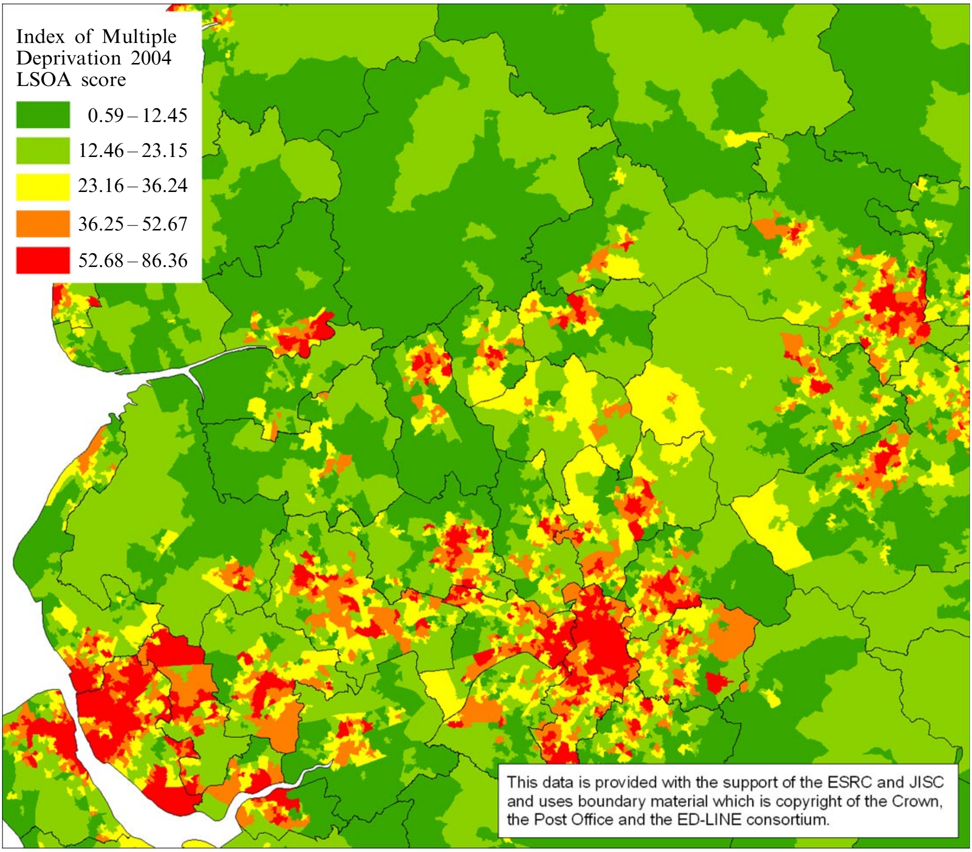


Figure 6. [In colour online.] Conceptualisation of spatial context (LSOA—lower super output area).

between rural and urban areas is much starker; the inner urban areas have the darkest shading and are generally most deprived in terms of their spatial context.

If we were to consider the level of deprivation in each LSOA only in relation to its own deprivation score, this would be to ignore its spatial context. For the purposes of deriving a typology based on origin – destination data, the intention is therefore to develop a formal measure of spatial context by recognising the potential spatial interactions with surrounding areas. The differences between the individual LSOA deprivation and the nearest-neighbour values is illustrated in table 4, which shows the fifteen LSOAs with the highest IMD deprivation scores in England, all of which happen to be in Merseyside or Manchester. Only four of these stay within the fifteen most deprived LSOAs in England in terms both of their individual LSOA score *and* of their nearest-neighbour LSOA score. In other cases, however, the ranking of wards can change quite radically. The implication is that areas with both a high individual deprivation score and a high nearest-neighbour score are much more likely to be classified as isolates by virtue of their spatial context since most household moves, especially in deprived areas, occur over short distances. Conversely, deprived LSOAs that are far removed from other deprived areas are more likely to have spatial interactions with different types of areas and have, in theory, greater potential for socioeconomic change as a result of migration patterns (see O'Reilly and Stevenson, 2003).



(a)

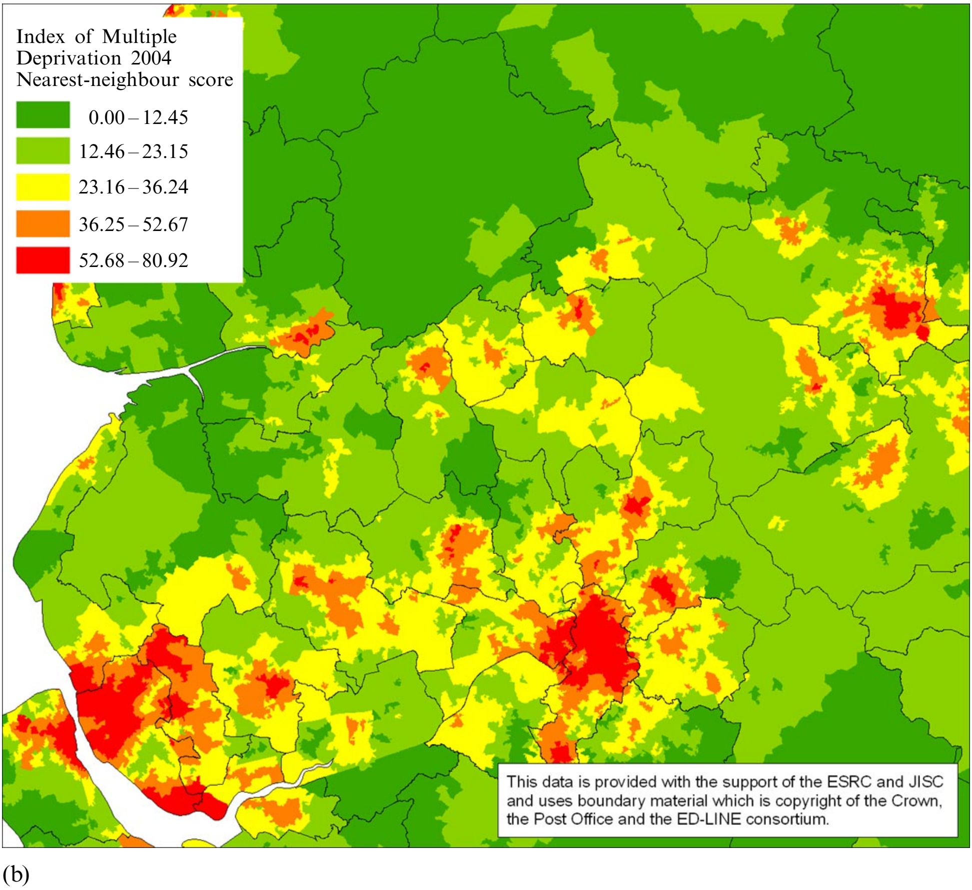
Figure 7. [In colour online.] (a) Lower super output area (LSOA) deprivation scores in northwest England; (b) nearest-neighbour deprivation scores in northwest England.

Ideally, developing a neighbourhood typology should take account of such contextual conditions by using nearest-neighbour values for each deprived LSOA across the country. This is a task which we will explore further in future research.

'Stayers'

Third is the important question of the 'stayers' within the LSOAs. Characterising neighbourhoods only in terms of those people who move runs the risk of ignoring the possibly very different characteristics of the immobile households. In other words, the movers may represent a through-flow of households with characteristics that differ from the 'stock' of relatively unchanging stayers. This would undoubtedly be a real problem if the focus was on the age of households since the propensity to move is so strongly associated with age. However, the principal focus of regeneration policy is on deprived households, and deprivation is not especially age-specific.[7] The issue of stayers might therefore centre on which of the four neighbourhood categories are likely to show radical differences between movers and stayers.

For the deprived LSOAs as a whole, the rate of residential mobility in 2000/01 was around 11.8%. As shown in table 5, there is some variation in churn rates between the four neighbourhood types, with improver and transit areas exhibiting slightly higher



(b)

Figure 7 (continued).

[7] Perhaps with the exception of the elderly poor, although even here there is no necessary correlation between poverty and old age (eg Bailey and Livingstone, 2007).

Table 4. Comparison of individual and nearest-neighbour Index of Multiple Deprivation (IMD) scores.

LSOA[a]	Ward	District	Individual LSOA		Nearest neighbour (NN)	
			score	rank	NN score	NN rank
E01006559	Breckfield	Liverpool	86.36	1	77.72	8
E01005204	Harpurhey	Manchester	85.76	2	79.85	3
E01006755	Speke	Liverpool	85.59	3	69.72	72
E01005133	Central	Manchester	84.92	4	70.85	59
E01005203	Harpurhey	Manchester	84.78	5	68.34	92
E01005067	Ardwick	Manchester	83.08	6	64.90	163
E01006468	Princess	Knowsley	82.30	7	78.54	6
E01006676	Granby	Liverpool	82.04	8	73.26	36
E01005202	Harpurhey	Manchester	81.89	9	71.16	54
E01006561	Breckfield	Liverpool	81.39	10	73.28	35
E01005106	Bradford	Manchester	81.26	11	72.21	43
E01005108	Bradford	Manchester	80.65	12	59.36	390
E01006778	Vauxhall	Liverpool	80.62	13	74.10	24
E01006469	Princess	Knowsley	80.49	14	65.67	143
E01006436	Kirkby Central	Knowsley	80.31	15	77.72	8

Note: data are for the 15 most deprived LSOAs as measured by individual IMD scores.
[a] LSOA—lower super output area.

churn rates than escalator and isolate areas. This is also evident from looking at the top and bottom quintile of the churn distribution. The neighbourhoods with the highest average levels of residential churn are improvers and transit areas, while isolate areas have the lowest levels. These differences are ones that would be expected from an interpretation of the roles of the different neighbourhood types.

Where households come from or move to neighbourhoods with different levels of deprivation (for example, the improver and escalator areas) the 'stayer' question may be less of an issue since the implication is that it is not just one subset of a neighbourhood population that moves. It may also not be a significant issue for the isolate areas since those moving in and those moving out are, by definition, largely deprived. The real issue may therefore be associated with the transit areas since both those moving in and those moving out are likely to be more affluent than those who remain in the neighbourhood. There is some indication that this may be case.

This can be explored by comparing the probability of movement of people who are more deprived or less deprived. The only robust data through which to analyse this is

Table 5. The neighbourhood typology and churn rates [source: Census, Table C0572, age (all people), and highest level of qualification (all people in households aged 25–74) by migration].

	Churn rate			Number of LSOAs[a]		
	average	minimum	maximum	total	low churn (bottom quintile)	high churn (top quintile)
Escalator	10.64	4.48	31.98	1213	304	149
Improver	13.08	4.57	61.30	521	90	129
Isolate	10.46	3.08	46.39	2030	522	220
Transit	13.17	4.70	71.95	2519	341	759

Note: churn is defined as the ratio of in-movers or out-movers (whichever is the higher) relative to the LSOA population.
[a] LSOA—lower super output area.

Table 6. Rates of residential out-moves: people with lower or higher qualifications [source: Census, Table C0572, age (all people), and highest level of qualification (all people in households aged 25 – 74) by migration].

Neighbourhood type	District classification	A: percentage of people with no/low qualifications who move out	B: percentage of people with higher qualifications who move out	C: percentage of all people who move out	Ratio A to C
Escalator	Conurbation core	6.2	12.5	8.0	0.78
	Conurbation industrial	6.6	11.5	7.8	0.84
	Industrial and mining	6.8	11.3	7.9	0.86
	Large free-standing city	6.5	11.9	7.8	0.83
	Large free-standing town	6.2	11.2	7.4	0.84
	London core districts	7.0	14.4	10.3	0.67
	London dormitory	7.6	12.1	9.1	0.84
	Non-London dormitory	6.4	9.9	7.3	0.88
	Rural and semirural	7.6	11.4	8.6	0.88
	Seaside resort	9.3	13.3	10.3	0.90
Escalator total		6.6	12.5	8.4	0.79
Improver	Conurbation core	7.7	15.7	10.2	0.75
	Conurbation industrial	6.6	11.5	7.8	0.85
	Industrial and mining	6.9	11.0	8.0	0.87
	Large free-standing city	6.9	13.8	8.9	0.78
	Large free-standing town	7.2	10.1	8.0	0.91
	London core districts	7.1	17.7	12.6	0.57
	London dormitory	6.5	12.4	8.5	0.77
	Non-London dormitory	6.3	9.7	7.2	0.87
	Rural and semirural	6.7	9.8	7.4	0.90
	Seaside resort	12.2	14.3	12.8	0.95
Improver total		7.1	14.5	9.5	0.75
Isolate	Conurbation core	6.6	11.9	7.9	0.83
	Conurbation industrial	6.6	11.1	7.6	0.86
	Industrial and mining	6.6	10.1	7.4	0.89
	Large free-standing city	6.5	11.8	7.7	0.84
	Large free-standing town	8.2	12.4	9.3	0.88
	London core districts	7.6	14.8	10.8	0.70
	London dormitory	7.2	10.2	8.1	0.89
	Non-London dormitory	7.2	10.6	8.0	0.89
	Rural and semirural	7.0	10.4	7.8	0.90
	Seaside resort	10.1	12.7	10.8	0.94
Isolate total		6.7	12.2	8.2	0.82
Transit	Conurbation core	7.0	17.2	10.6	0.66
	Conurbation industrial	7.1	13.0	8.7	0.81
	Industrial and mining	7.0	11.8	8.2	0.86
	Large free-standing city	7.9	17.5	11.2	0.71
	Large free-standing town	7.5	15.4	10.0	0.75
	London core districts	7.4	17.2	12.3	0.60
	London dormitory	8.1	13.1	9.7	0.83
	Non-London dormitory	6.7	12.2	8.3	0.81
	Rural and semirural	8.1	13.1	9.5	0.85
	Seaside resort	10.8	15.7	12.4	0.87
Transit total		7.6	15.2	10.1	0.74

census information on the educational qualifications of individuals.[8] People who are 'more deprived' are taken as those with no qualifications or only level 1; the less deprived are taken as those with level 2 qualifications or above.[9]

Table 6 shows these movement probabilities (expressed as percentages of those with lower qualifications and with higher qualifications, and the total population who moved between LSOAs during 2000 and 2001) distinguishing the fourfold neighbourhood typology and the tenfold district classification.

A number of significant conclusions can be drawn from these data. First, the highest overall percentages of mobility are found for the transit and improver neighbourhoods, and this is predominantly the result of high percentages of moves involving better-qualified people. Equally significant is the fact that the lowest overall churn is linked to isolate areas. This adds valuable support to the robustness of the neighbourhood categories. Second, London and to a lesser extent the conurbation cores have very high rates of churn amongst the better qualified. While this is especially true for transit and improver areas, London stands out in this respect across all four types of neighbourhood. This distinctiveness reflects the very different housing market in London (and to a lesser extent the big conurbations). Third—and directly relevant to the issue of 'stayers'—there is relatively little difference between the rates of churn for less-qualified people in different types of district or different types of neighbourhood. With the exception of seaside resorts, the percentages vary only between roughly 6.0% and 8.0%. Indeed, of the three cases where percentages are above 8.0, two are in transit areas and one is in an isolate area. Had it been the case that transit areas had a significant 'trapped' population, one would have expected fewer of the deprived people in such areas to be residentially mobile. To this extent, the 'stayer' problem may be less of an issue than at first appears.

Nevertheless, especially in London, the large differences between high levels of mobility of the better-qualified and the lower levels for the less-qualified do imply that the classifications of neighbourhood types (based as they are on those who move) need further exploration, especially in the case of areas in the London core. Again, this is a question that will be explored by further research.

Conclusion

There is clearly a need for greater understanding of the different roles that neighbourhoods play. Interpreting the rich data from the migration tables of the 2001 Census provides some evidence of the variation between areas in relation to their role in the ebb and flow of residential mobility. The fourfold typology of neighbourhood roles explored in this paper is given considerable support in the empirical patterns suggested in figures 3 to 5. The four neighbourhood types make considerable sense in terms of knowledge of neighbourhoods in the respective areas. The different types of neighbourhood are not simply a function of the degree of deprivation in each of the neighbourhoods; rather, they appear to point to real differences in the differing functional roles that different neighbourhoods play. This suggests that the fourfold typology suggested here provides a useful platform on which to base a categorisation of neighbourhood types.

[8] The data cover people aged 25 years and over so as to exclude the distorting effects of high churn in areas with large student populations. Educational data have been used since this is the only individual characteristic covered by the census that is unlikely to have changed in the year prior to the census.

[9] Level 1 qualifications are one or more O-level, CSE, or GSCE; a level 2 qualification is a Foundation GNVQ.

The evidence, however, is as much tantalising as definitive, not least since there are still technical and conceptual conundrums that further research will need to explore. Our intention is to pursue two of these further: the issue of incorporating a measure of deprivation based on nearest-neighbour averages in order to take account of the wider spatial context in which deprived areas sit; and how best to incorporate the issue of 'stayers' into the interpretation of neighbourhood types, especially in the case of London.

In the longer term, the applied relevance of a neighbourhood typology such as this is that a consistent and robust set of neighbourhood types has obvious implications both for the development and for the evaluation of neighbourhood-based regeneration policy. This study is part of a wider evaluation of the National Strategy for Neighbourhood Renewal,[10] and the policy implications of the work discussed here will eventually be reported to the sponsor, the Department for Communities and Local Government.

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[10] The overall project is led by Amion Consulting Ltd.

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